

12.3 Latent Diffusion and Text Conditioning

12× fewer latent cells. CFG: $s=1$ is still conditional.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

Pixel-space diffusion on 256×256 RGB is expensive. **Latent diffusion** (Rombach et al.) denoises in a compressed space. **Text** enters as a condition, usually from a frozen CLIP text tower (Week 5). By the end you should be able to count latent cells against pixels, write classifier-free guidance, and say what $s = 1$ is not.

1. What latent diffusion and classifier-free guidance are

Latent diffusion denoises in a **VAE latent**, not in pixels. A pretrained encoder E maps an image to $z_0 = E(x)$. You run the forward process of note **12.1** on z , not on pixels. A decoder D_{VAE} turns the final latent back into an image. Training the U-Net on latents is cheaper; the VAE already threw away some high-frequency detail.

Classifier-free guidance (CFG) is how you steer that denoiser with a caption. You train one net with and without the text, then at sample time mix a conditional noise prediction ε_c with an unconditional prediction ε_u . Scale s sharpens the condition. $s = 1$ is still conditional (it is just the conditional net). $s = 0$ is the uncond branch. High s is sharper and less diverse.

Text is a CLIP (or T5) vector. CLIP scores pairs; it does not decode a caption (note **5.2**). Here you only need the **text** embedding as a handle. You want $p(x \mid \text{caption})$, not an unconditional generator.

2. Why we use it

Pixels are huge. Latent cells are fewer (on the order of $12 \times$ fewer in the classroom cartoon, about $48 \times$ on Stable Diffusion's public $64 \times 64 \times 4$ for 512^2). Fifty reverse steps on the small tensor are cheaper than fifty steps on pixels.

Text needs a condition or you sample generic images. Concatenating a CLIP vector, or using **cross-attention** so latent tokens query text tokens, is how the caption enters the U-Net. CFG exists because a single conditional prediction is often too mild: extrapolating away from the unconditioned guess is a cheap way to raise prompt match, at the cost of variety.

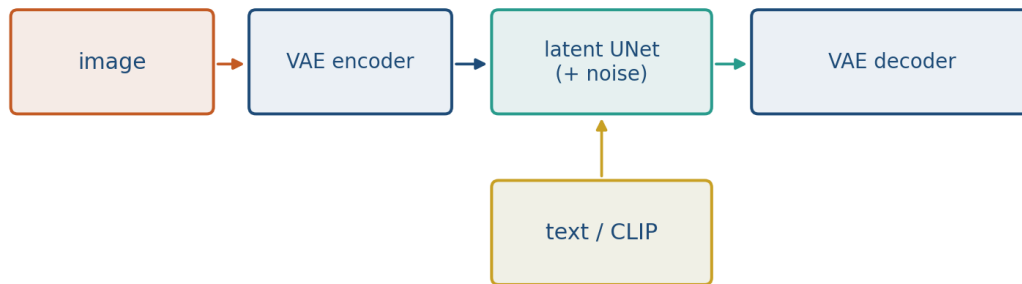
Conditioning does not remove Week 5's bias story: the same CLIP geometry is in the loop.

3. Architecture

Three frozen or trained pieces sit around the denoiser you already know.

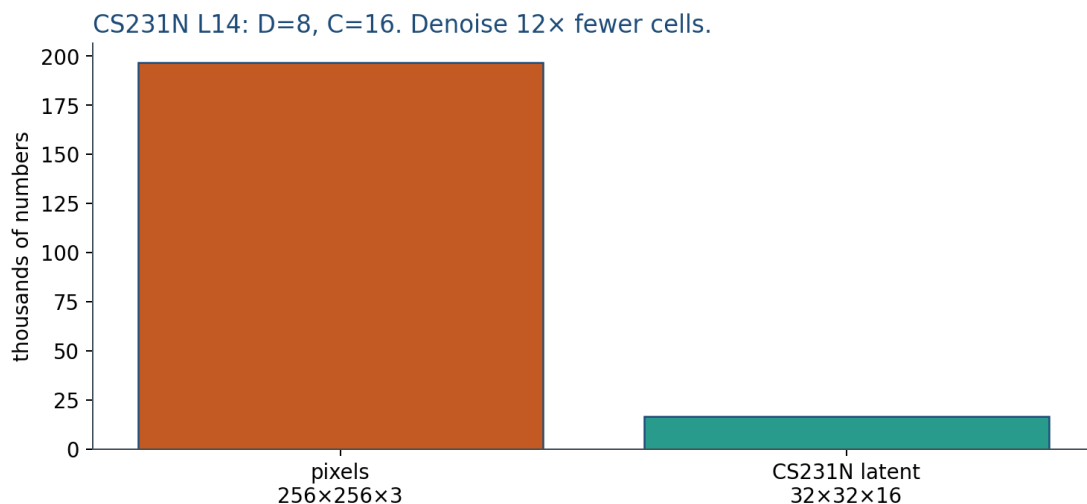
1. **VAE encoder / decoder.** Frozen at diffusion-training time in the usual recipe. Encode $x_0 \rightarrow z_0$, diffuse z , decode at the end.
2. **Denoiser on z_t .** U-Net or DiT, with a time embedding, now also taking a condition c .
3. **Text encoder.** CLIP's text tower (note **5.1**) or T5. Inject c by concatenating it to the timestep embedding, or by **cross-attention**.

Latent diffusion: denoise a cheap latent, condition on text



Stable Diffusion is this picture plus a big text-conditioned U-Net. The idea is the same at homework scale: compress, denoise, decode.

Typical sizes to say out loud: an image $512 \times 512 \times 3$ becomes a latent like $64 \times 64 \times 4$. You pay diffusion steps on the small tensor. A classroom setting is $D = 8, C = 16$: $256 \times 256 \times 3 = 196,608$ numbers versus $32 \times 32 \times 16 = 16,384$ (about $12 \times$ fewer). Stable Diffusion's public $64 \times 64 \times 4$ on 512^2 is about $48 \times$. Denoise the small tensor.



Modern LDM pipelines can stack a VAE, a GAN (for the decoder), and diffusion. DiT exists. This hour is the VAE-latent picture, not a transformer-block homework. CFG is two forwards at sample time, then a mix: $\hat{\epsilon} = \epsilon_u + s(\epsilon_c - \epsilon_u)$.

4. How it works, step by step

Latent path.

1. Encode the image: $z_0 = E(x)$.
2. Run the forward noising of note **12.1** on z , train the reverse of note **12.2** on z_t , conditioned on c when a caption is present.
3. At sample time, start from noise in latent space, walk the reverse chain, decode $\hat{x} = D_{\text{VAE}}(z_0)$.

Conditioning. Encode the caption with CLIP's text tower. During training, **randomly drop** the text so one net is both conditional and unconditional (the empty prompt $\emptyset$).

Guidance at sample time.

1. Run the net twice (or batch the two): $\varepsilon_u = \varepsilon_\theta(z_t, t, \emptyset)$ and $\varepsilon_c = \varepsilon_\theta(z_t, t, c)$.
2. Mix with scale s : $\hat{\varepsilon} = \varepsilon_u + s(\varepsilon_c - \varepsilon_u)$.
3. Plug $\hat{\varepsilon}$ into the reverse-step mean from note **12.2**.

$s = 1$ recovers ε_c : still conditional, not "off." $s > 1$ **extrapolates** away from the unconditioned guess. A huge s can look like a poster of the prompt and still miss modes (Week 11's coverage lesson, now in prompt space). Report s .

The lab for this week is still 2-D denoising from notes **12.1–12.2**. This note is the picture that products use; you do not train Stable Diffusion here.

5. Mathematical formulas

Classifier-free guidance mixes two noise predictions:

$$\hat{\varepsilon}_\theta = \varepsilon_\theta(z_t, t, \emptyset) + s(\varepsilon_\theta(z_t, t, c) - \varepsilon_\theta(z_t, t, \emptyset)).$$

At $s = 1$, $\hat{\varepsilon}_\theta = \varepsilon_\theta(z_t, t, c)$. At $s = 0$, you are on the unconditional branch. The latent forward process is the same Gaussian jump as note **12.1**, with z in place of x .

6. Positive points and negative points

Positive.

- Denoising a latent is cheaper than pixel DDPM by a large constant (classroom $\sim 12\times$, public SD $\sim 48\times$).
- A frozen CLIP text tower is enough to steer generation without decoding a caption.
- CFG is one scalar s that raises prompt match without training a separate classifier.
- Randomly dropping text in training gives you both branches from one net.

Negative.

- The VAE throws away high-frequency detail; decode artifacts are part of the method.
- CFG is a quality–diversity knob: high s sharpens and washes out variety.
- You still pay many reverse steps, now twice per step if you run both branches.
- CLIP’s image tower is for scoring pairs, not for sampling; fine-tuning it at sample time is the wrong tower.

When not to. Unconditional 2-D denoising (Lab 12) does not need a VAE or CFG. A task that only needs a CLIP similarity score does not need a decoder.

7. What to measure

Text-to-image is not “the picture looks nice.” Report a prompt-alignment score (CLIP cosine of image vs text), a coverage or diversity check, and a failure case (wrong count, wrong object, reading text in the image). Conditioning does not remove Week 5’s bias story: the same CLIP geometry is in the loop.

8. Teaching this note

About **35 minutes** at the board, then **~12 minutes** of selected video from the first 45 minutes (do not play the five-hour coding rest).

- **0–12 min.** Why latents: VAE encode / denoise / decode. Count pixels vs latent cells.
 - **12–22 min.** CLIP text tower as c . Cross-attention in one sentence.
 - **22–33 min.** Classifier-free guidance with numbers. Diversity vs prompt match.
 - **Then** play Umar Jamil Stable Diffusion **4:30–12:07** (what SD is; forward/reverse in this stack) and **22:20–31:00** (CFG). Optional 31:00–39:54 (CLIP + VAE + text-to-image) if time. Stop by **45:00**.
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9. Worked example

Latent size. A $512 \times 512 \times 3$ image is 786,432 numbers. A $64 \times 64 \times 4$ latent is 16,384 numbers: about $48 \times$ fewer. Fifty reverse steps on the latent are cheaper than fifty steps on pixels.

Guidance. Let the net’s two noise guesses (one scalar, to see the algebra) be $\varepsilon_{\theta}(z_t, t, \emptyset) = 0.8$ (unconditional) and $\varepsilon_{\theta}(z_t, t, c) = 0.2$ (with prompt c). The difference is $0.2 - 0.8 = -0.6$.

$$\hat{\varepsilon} = 0.8 + s(-0.6) = 0.8 - 0.6s.$$

s	$\hat{\epsilon}$	reading
1	0.2	just the conditional net
3	-1.0	step farther along the prompt direction
7.5	-3.7	very sharp prompt match; variety usually drops

$s = 1$ is “use the text-conditioned prediction.” $s > 1$ **extrapolates** away from the unconditioned guess. That is why a huge s can look like a poster of the prompt and still miss modes (Week 11’s coverage lesson, now in prompt space).

 CFG scale <math xmlns=

$s=1,3,7.5$ on a scalar pair" src="graphics/12.3-latent-conditioning/cfg-scale.png" />

Randomly **drop** the text in training so one net is both conditional and unconditional. Two forwards at sample time, then mix with s .

10. Where students get stuck

- Fine-tuning CLIP’s image tower at sample time. Text-to-image needs the **text** tower (and a VAE decoder). The image tower is for scoring pairs, not for sampling.
- Calling $s = 1$ “no text.” $s = 1$ is still conditional; the empty prompt $\emptyset$ is the other call.
- Measuring only FID or only “looks nice,” with no prompt-alignment number and no failure case.

11. Video

Watch [Umar Jamil, Coding Stable Diffusion from scratch in PyTorch](#), **0:00–45:00** assigned, with in-class pauses at **4:30–12:07** (what Stable Diffusion is; forward/reverse) and **22:20–39:54** (classifier-free guidance, CLIP, VAE, text-to-image).

Do not start the coding chapters (VAE implementation from 44:30 onward) in this lecture. The board is the picture; the lab is still 2-D denoising from notes **12.1–12.2**.

12. Practice

1. Why denoise latents instead of pixels if you already have a VAE?
2. CLIP’s image tower is optional at sample time for text-to-image. Which CLIP tower is not?
3. Guidance scale $s = 1$ versus $s = 12$: what do you expect to happen to prompt match and to variety?
4. $\epsilon_{\text{uncond}} = 1.0$, $\epsilon_{\text{cond}} = 0.4$, $s = 5$. Compute $\hat{\epsilon}$.

5. A $256 \times 256 \times 3$ image vs an $32 \times 32 \times 4$ latent: how many numbers in each, and what is the ratio (image / latent)?